Question 2

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```
data = read.csv("Admissions.csv")
head(data)
```

```
##      Admit Gender Dept Freq
## 1 Admitted   Male    A  512
## 2 Rejected   Male    A  313
## 3 Admitted Female    A   89
## 4 Rejected Female    A   19
## 5 Admitted   Male    B  353
## 6 Rejected   Male    B  207
```

a)

```
tab = xtabs(Freq ~ Gender + Admit, data=data)
tab
```

```
##         Admit
## Gender   Admitted Rejected
##   Female      557     1278
##   Male       1198     1493
```

```
prop_tab = prop.table(tab, margin=1)
prop_tab
```

```
##         Admit
## Gender    Admitted  Rejected
##   Female 0.3035422 0.6964578
##   Male   0.4451877 0.5548123
```

The 2×2 table shows that there are considerably more male applicants (2691) than female applicants (1835), suggesting that men applied in greater numbers overall. Looking at the admission proportions, men were admitted at a rate of 44.5% compared to only 30.4% for women, a difference of approximately 14 percentage points. This suggests a substantial disparity in admission rates by gender, with men being notably more likely to be admitted than women.

b)

```
prop.test(x= tab[,1], n=c(tab[1,1]+ tab[1, 2],tab[2,1]+ tab[2, 2]), correct=FALSE)
```

```
##
##  2-sample test for equality of proportions without continuity correction
##
## data:  tab[, 1] out of c(tab[1, 1] + tab[1, 2], tab[2, 1] + tab[2, 2])
## X-squared = 92.205, df = 1, p-value < 2.2e-16
## alternative hypothesis: two.sided
## 95 percent confidence interval:
##  -0.1698439 -0.1134470
## sample estimates:
##    prop 1    prop 2
## 0.3035422 0.4451877
```

Conclusion: The two-sample proportion test yields a test statistic of $X^2 = 92.205$ with 1 degree of freedom and a p-value < 2.2e-16, which is far below 0.05. Therefore we reject the null hypothesis of equal admission probabilities between genders. There is highly significant evidence that the probability of admission differs between men and women — with women admitted at a rate of 30.4% compared to 44.5% for men, a difference of approximately 14 percentage points (95% CI: [-0.170, -0.113]).

c)

```
model.c = glm(Freq~., data=data, family="poisson")
summ.c = summary(model.c)
summ.c
```

```
##
## Call:
## glm(formula = Freq ~ ., family = "poisson", data = data)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)    4.98824    0.04177 119.424  < 2e-16 ***
## AdmitRejected  0.45674    0.03051  14.972  < 2e-16 ***
## GenderMale     0.38287    0.03027  12.647  < 2e-16 ***
## DeptB         -0.46679    0.05274  -8.852  < 2e-16 ***
## DeptC         -0.01621    0.04649  -0.349 0.727355
## DeptD         -0.16384    0.04832  -3.391 0.000696 ***
## DeptE         -0.46850    0.05276  -8.879  < 2e-16 ***
## DeptF         -0.26752    0.04972  -5.380 7.44e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 2650.1  on 23  degrees of freedom
## Residual deviance: 2097.7  on 16  degrees of freedom
```

```
## AIC: 2272.7
##
## Number of Fisher Scoring iterations: 5
```

Interpretation of parameters:

Intercept (4.988): The log expected count for the baseline group (Admitted, Female, Dept A). AdmitRejected (0.457): Being rejected increases the log count by 0.457 relative to being admitted, holding other variables constant. GenderMale (0.383): Being male increases the log count by 0.383 relative to female, meaning men have higher counts across admission categories. DeptB to DeptF: Each coefficient represents the log count difference relative to Dept A. Departments B and E have the largest negative effects.

```
pchisq(model.c$deviance, df=model.c$df.residual, lower.tail=FALSE)
```

```
## [1] 0
```

The p-value of $0 < 0.05$ means the residual deviance of 2097.7 on 16 degrees of freedom is extremely large. Therefore we reject the null hypothesis of independence between admission status and gender. There is highly significant evidence that admission status and gender are not independent the probability of admission differs significantly by gender.

d)

```
tab.d = xtabs(Freq ~ Gender + Admit + Dept, data=data)
tab.d
```

```
## , , Dept = A
##
##         Admit
## Gender   Admitted Rejected
##   Female       89       19
##   Male        512      313
##
## , , Dept = B
##
##         Admit
## Gender   Admitted Rejected
##   Female       17        8
##   Male        353      207
##
## , , Dept = C
##
##         Admit
## Gender   Admitted Rejected
##   Female      202      391
##   Male        120      205
##
## , , Dept = D
##
##         Admit
```

```
## Gender   Admitted Rejected
##   Female      131      244
##   Male        138      279
##
## , , Dept = E
##
##         Admit
## Gender   Admitted Rejected
##   Female       94      299
##   Male         53      138
##
## , , Dept = F
##
##         Admit
## Gender   Admitted Rejected
##   Female       24      317
##   Male         22      351
```

```
dept_tab = prop.table(tab.d, margin=c(1,3))
dept_tab
```

```
## , , Dept = A
##
##         Admit
## Gender     Admitted   Rejected
##   Female 0.82407407 0.17592593
##   Male   0.62060606 0.37939394
##
## , , Dept = B
##
##         Admit
## Gender     Admitted   Rejected
##   Female 0.68000000 0.32000000
##   Male   0.63035714 0.36964286
##
## , , Dept = C
##
##         Admit
## Gender     Admitted   Rejected
##   Female 0.34064081 0.65935919
##   Male   0.36923077 0.63076923
##
## , , Dept = D
##
##         Admit
## Gender     Admitted   Rejected
##   Female 0.34933333 0.65066667
##   Male   0.33093525 0.66906475
##
## , , Dept = E
##
##         Admit
## Gender     Admitted   Rejected
##   Female 0.23918575 0.76081425
```

```
##   Male   0.27748691 0.72251309
##
## , , Dept = F
##
##         Admit
## Gender     Admitted   Rejected
##   Female 0.07038123 0.92961877
##   Male   0.05898123 0.94101877
```

```r
for(dept in c("A", "B", "C", "D", "E", "F")){
  # subset data for this department
  dept_data = subset(data, Dept == dept)

  # independence model
  model_ind = glm(Freq ~ Gender + Admit, data=dept_data, family=poisson)

  # test via deviance
  pval = pchisq(model_ind$deviance, df=model_ind$df.residual, lower.tail=FALSE)

  cat("Dept", dept, "- Deviance:", round(model_ind$deviance, 3),
      "- p-value:", round(pval, 4), "\n")
}
```

```
## Dept A - Deviance: 19.054 - p-value: 0
## Dept B - Deviance: 0.259 - p-value: 0.6111
## Dept C - Deviance: 0.751 - p-value: 0.3862
## Dept D - Deviance: 0.298 - p-value: 0.5852
## Dept E - Deviance: 0.99 - p-value: 0.3196
## Dept F - Deviance: 0.384 - p-value: 0.5357
```

Conclusion: While the previous result rejected automatically, for the department-level analysis we have the following results:

For departments B through F, the p-values are all greater than 0.05, so we fail to reject the null hypothesis of independence there is no significant evidence that admission status depends on gender within these departments. However, in department A yields a p-value = 0, so we reject independence gender significantly affects admission in department A, where women are admitted at a notably higher rate (82.4%) than men (62.1%).

e)

```r
results = data.frame(Dept=character(), Accept_Rate=numeric(), Female_Prop=numeric())

for(dept in c("A", "B", "C", "D", "E", "F")){
  dept_data = subset(data, Dept == dept)

  total = sum(dept_data$Freq)
  admitted = sum(dept_data$Freq[dept_data$Admit == "Admitted"])
  female = sum(dept_data$Freq[dept_data$Gender == "Female"])

  accept_rate = admitted / total
```

```
  female_prop = female / total

  results = rbind(results, data.frame(
    Dept = dept,
    Accept_Rate = round(accept_rate, 3),
    Female_Prop = round(female_prop, 3)
  ))
}

print(results)
```

```
##   Dept Accept_Rate Female_Prop
## 1    A       0.644       0.116
## 2    B       0.632       0.043
## 3    C       0.351       0.646
## 4    D       0.340       0.473
## 5    E       0.252       0.673
## 6    F       0.064       0.478
```

These numbers shed considerable light on the cause of the paradox. Departments A and B have the highest overall acceptance rates (64.4% and 63.2%) but the lowest proportions of female applicants (11.6% and 4.3%). Conversely, departments C, E and F, which have the lowest acceptance rates (35.1%, 25.2% and 6.4%), have the highest proportions of female applicants (64.6%, 67.3% and 47.8%).

Conclusion

The results yield that women disproportionately apply to the most competitive and selective departments, while men concentrate their applications in the most accessible ones. As a result, even though women are admitted at comparable or higher rates than men within each department, their overall admission rate appears lower in the aggregate which leads to the Simpson Paradox observed.